

Takumi Watanabe

AI Engineer | Agentic RAG & Production LLM Systems

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PROFILE

Software Engineer with 13 years of software engineering experience, currently specializing in production LLM and RAG systems using Python, LangChain, and AWS. Shipped Agentic RAG systems from concept to production achieving 30% accuracy improvement and sub-second latency. Expert in agentic workflows, hybrid search, vector databases, Docker, and Kubernetes with full-stack ownership.

EDUCATION

Master of Science in Computer Science | California State University, Long Beach | December 2012

Bachelor of Science in Mathematics | California State University, Long Beach | May 2011

CERTIFICATIONS

AWS Certified Generative AI - Professional | 2026

AWS Certified Machine Learning - Specialty | 2026

TECHNICAL SKILLS

AI & Machine Learning:	LLM, RAG, Agentic Workflows, LangChain, Vector Databases (Qdrant), Ollama
LLM Evaluation:	RAGAS, LangSmith, Prompt Engineering, A/B Testing, LLM Tracing
Backend Development:	Python (FastAPI, Pydantic, Celery), SQL (PostgreSQL), REST APIs
Cloud & Infrastructure:	AWS (Lambda, ECS, S3, DynamoDB), Docker, Kubernetes, Terraform, CI/CD
Data Engineering:	Hybrid Search, Embeddings, BM25, RRF Fusion, Query Planning, HNSW

WORK EXPERIENCE

Principal Engineer

GenArchitect, Remote

July 2023 - Present

- Deployed enterprise RAG solution processing 10K+ documents with continuous evaluation and benchmarking, achieving 30% accuracy improvement over baseline and sub-second response times.
- Established gold-standard evaluation dataset using RAGAS metrics (Faithfulness, Answer Relevance) enabling data-driven deployments with quantifiable accuracy benchmarks.
- Architected agentic AI system with tool-calling, multi-turn reasoning, and context management enabling autonomous task completion and intelligent user interactions.
- Integrated local LLM inference with Ollama for privacy-first answer generation, eliminating external API costs while ensuring data security compliance for sensitive document workflows.

Principal Cloud Software Engineer

Cylance / BlackBerry, Irvine, CA

May 2019 - July 2023

- Built scalable Python microservices deployed to 6 AWS production regions serving 14M+ endpoints with 99.9% uptime for critical incidents and real-time security monitoring.

- Architected threat intelligence system processing 4M monthly event feeds using AWS Kinesis, Lambda, Kafka, and Elasticsearch for real-time security analytics serving enterprise customers globally.
- Reduced AWS infrastructure costs by 83% from \$300K to \$50K monthly (\$250K annual savings) through event processing architecture redesign with cost-optimized data streams and auto-scaling.
- Automated infrastructure provisioning for 14 AWS resource types using Terraform IaC and Jenkins CI/CD reducing deployment time by 70% across 6 production regions.

Cloud Engineer Consultant

Microsoft (via Collabera), Remote, WA

January 2022 - June 2022

- Modernized 2 legacy services refactoring 15 deeply nested object models into .NET Core microservices optimized for Kubernetes deployment with container orchestration and auto-scaling.
- Enhanced reliability and scalability of systems powering order fulfillment and financial transactions processing \$50M+ annual revenue on Microsoft internal marketplace.

Principal Software Consultant

Bio-Rad (via Technossus), Irvine, CA

January 2017 - May 2019

- Developed 6 microservice REST APIs using Python and Node.js in AWS reducing infrastructure costs by 40%.
- Implemented OAuth 2.0, JWT authentication, and RBAC across 15 microservices achieving SOC 2 compliance.

PROFESSIONAL PROJECTS

RAGnosis | <https://www.ragnosis.app/chat>

2026 - Ongoing

Production agentic RAG system with query planning, hybrid search, and automated quality evaluation

- Architected agentic RAG pipeline with query planning, semantic expansion, and automated evaluation achieving 0.91 faithfulness score via RAGAS framework with 28 test questions.
- Implemented hybrid search combining vector (pgvector HNSW) and keyword search (BM25) with RRF fusion (60/40 weighting) improving accuracy over naive semantic search.
- Mitigated context window overload and "lost in the middle" degradation through query-aware chunk selection and relevance-based context pruning, maintaining accuracy at scale.

On-Prem RAG

2025 - 2026

Production-grade retrieval-augmented generation system with hybrid search (70% semantic + 30% BM25), cross-encoder reranking, and local LLM integration for intelligent document querying

- Architected scalable RAG pipeline with FastAPI, Celery workers, Qdrant vector database, PostgreSQL metadata storage, and Redis message broker serving real-time queries.
- Implemented hybrid retrieval combining semantic vector search (fastembed 384D) and BM25 keyword search with 70/30 weighted fusion plus cross-encoder reranking for improved accuracy.
- Integrated local LLM inference with Ollama for context-aware answer generation, eliminating API costs and ensuring data privacy through on-premise processing.

CORE COMPETENCIES

Full-Stack System Ownership • Data-Driven Decision Making • Cross-Functional Collaboration • Rapid Prototyping & Iteration • Performance & Cost Optimization • Technical Problem Solving • Production Reliability Engineering • Technical Mentorship & Knowledge Sharing • Continuous Learning & Innovation